

Anthraquinones Inhibit Tau Aggregation and Dissolve Alzheimer's Paired Helical Filaments *in Vitro* and in Cells*

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The abnormal aggregation of tau protein into paired helical filaments (PHFs) is one of the hallmarks of Alzheimer's disease. Aggregation takes place in the cytoplasm and could therefore be cytotoxic for neurons. To find inhibitors of PHF aggregation we screened a library of 200,000 compounds. The hits found in the PHF inhibition assay were also tested for their ability to disassemble preformed PHFs. The results were obtained using a thioflavin S fluorescence assay for the detection and quantification of tau aggregation in solution, a tryptophan fluorescence assay using tryptophan-containing mutants of tau, and confirmed by a pelleting assay and electron microscopy of the products. Here we demonstrate the feasibility of the approach with several compounds from the family of anthraquinones, including emodin, daunorubicin, adriamycin, and others. They were able to inhibit PHF formation with IC_{50} values of 1–5 μM and to disassemble preformed PHFs at DC_{50} values of 2–4 μM . The compounds had a similar activity for PHFs made from different tau isoforms and constructs. The compounds did not interfere with the stabilization of microtubules by tau. Tau-inducible neuroblastoma cells showed the formation of tau aggregates and concomitant cytotoxicity, which could be prevented by inhibitors. Thus, small molecule inhibitors could provide a basis for the development of tools for the treatment of tau pathology in AD and other tauopathies.

In Alzheimer's disease (AD)¹ two main proteins form abnormal polymers (1). The extracellular amyloid plaques consist largely of polymerized $A\beta$ -peptide, whereas the intracellular neurofibrillary tangles are made from the microtubule-associated protein tau (2). These insoluble aggregates or their oligomeric precursors are involved in neuronal degeneration. The distribution of the neurofibrillary changes can be used for the diagnosis and staging of AD (3); it is based on the appearance of tau in an aggregated state that in addition is chemically modified (phosphorylation, truncation, and glycation). Whether these modifications are the cause or merely by-products of neuronal degeneration has been a matter of debate. For example, different kinase pathways have been considered responsible for the early stages of neurodegeneration (4, 5). On the

other hand, unphosphorylated tau can be induced to aggregate, and the phosphorylation of tau retards aggregation *in vitro* (6). Examples from other protein aggregation diseases suggest that an increase in concentration drives the protein into aggregation and causes toxic effects (7), and conversely the reduction of aggregates alleviates the diseases (8, 9). In the case of $A\beta$ aggregation in AD, cytotoxicity may be caused by the extracellular plaques or by prefibrillary oligomers within cells (10). Since the discovery of inherited tau pathologies (FTDP-17) various cell models, transgenic mice, and other organisms have been generated to study tau aggregation (11, 12), but they do not yet reflect the full spectrum of the human pathology, and there is a debate on which form of tau is responsible for its toxicity.

Evidence for cytotoxicity of intracellular aggregates comes from other neurodegenerative diseases like Parkinson's and Huntington's disease (13, 14). In Parkinson's disease the cytotoxicity of α -synuclein has been traced back to prefibrillary oligomers binding to membranes (15). In Huntington's disease aggregated protein can be found in the nucleus, possibly affecting gene transcription. In cellular and mouse models the disaggregation of polymers improved viability (9, 16).

In the case of tau we face the paradoxon that the protein is highly soluble yet can aggregate into insoluble polymers. Soluble tau is a natively unfolded protein with mostly random coil conformation (17–19), but certain motifs can undergo a conformational change toward β -structure, which promotes the formation of AD-like filaments. Because the aggregation of tau in AD correlates with the clinical progression of the disease, it seemed likely that inhibition or reversal of tau aggregation could protect the affected neurons. To test this idea we began a search for inhibitors of tau aggregation and screened a library of 200,000 compounds. 1266 compounds were positive; 77 were even capable of dissolving preformed aggregates at low micromolar concentrations, and a subclass is described here. The compounds do not impair the physiological function of tau of stabilizing microtubules. Finally, cell models that show tau aggregation and cytotoxicity after inducible expression of tau can be rescued by inhibitors of aggregation.

MATERIALS AND METHODS

Chemicals and Proteins—Heparin (average molecular weight of 3000), polyglutamate (average molecular weight of 600 or 1000), and thioflavin S were obtained from Sigma. Full-length tau isoforms htau23 and htau24 and constructs of the repeat domain of tau (see Fig. 1B) were expressed in *Escherichia coli* and purified by making use of the heat stability and fast protein liquid chromatography Mono S (Amersham Biosciences) chromatography as described (20). Emodin, daunorubicin, and adriamycin were obtained from Merck. PHF016 was obtained from ChemBridge, and PHF005 was obtained from Interchim. All of the experiments presented here were carried out with freshly dissolved compounds.

PHF Formation *in Vitro*—Assembly of PHFs from tau protein (K19, 10 μM) was performed at 37 °C in the presence of polyanions (heparin,

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¹ The abbreviations used are: AD, Alzheimer's disease; PHF, paired helical filaments; ThS, thioflavin S; PBS, phosphate-buffered saline; PIPES, 1,4-piperazinediethanesulfonic acid.

2.5 μM) in 50 mM NH_4Ac , pH 6.8. Assembly was followed either qualitatively by electron microscopy or quantitatively by fluorescence assay using thioflavin S. PHF formation from tau isoforms htau23 and htau24 was carried out in PBS buffer, pH 7.4, 10 μM protein, and 2.5 μM heparin. The samples were incubated at 50 °C for 10 days. In the case of hTau24 and K18, dithiothreitol was added at a final concentration of 1 mM each day to avoid intramolecular disulfide cross-linking (21).

Screening of Tau Aggregation Inhibitors by Thioflavin S Assay—PHF formation was monitored by a thioflavin S fluorescence assay (22). Using an automated pipetting system (Cybi-Well; CyBio, Jena, Germany) 50 mM NH_4Ac , 10 μM protein (K19), 60 μM compound, and 2.5 μM heparin were mixed in 50 μl of volume in a 384-well plate (black microtiter 384 plate round well; ThermoLabsystems, Dreieich, Germany) and incubated overnight at 37 °C. As a control the protein was replaced with H_2O to measure the fluorescence of the compounds. After incubation with the compounds, thioflavin S was added to a final concentration of 20 μM , and the signal was measured at excitation of 440 nm and emission of 521 nm in a spectrofluorimeter (Ascent; Lab-systems, Frankfurt, Germany). To measure the inhibition of PHF assembly and disassembly of PHFs, we chose compound concentrations of 200, 60, and 10 μM down to 10 pM at a 10 μM concentration of tau. All of the measurements were carried out three times.

Tryptophan Fluorescence Spectroscopy—The fluorescence experiments were performed on a Spex Fluoromax spectrophotometer (Plytec, Waldbronn, Germany) using $3 \times 3\text{-mm}$ microcuvettes from Hellma (Mühlheim, Germany) with 20- μl sample volumes. Tryptophan emission was scanned from 300 to 450 nm at fixed excitation wavelength of 290 nm. For the inhibition assay, 60 μM compounds were incubated with constructs K19/Y310W, K18/Y310W, or K18 Δ K280/Y310W (10 μM) and heparin (2.5 μM) in PBS, pH 7.4, for 3 days at 37 °C. Dithiothreitol was added at a final concentration of 1 mM in the case of K18/Y310W and K18 Δ K280/Y310W. For the disassembly assay, 60 μM inhibitor compound were added to preformed PHFs (10 μM) and incubated overnight at 37 °C. PHFs were formed by incubation of tau construct K19/Y310W, K18/Y310W, or K18 Δ K280/Y310W (10 μM) with 2.5 μM heparin in volume of 100 μl at 37 °C in PBS, pH 7.4. Dithiothreitol at final concentration 1 mM was added in the case of the two last constructs. Incubation time was 3 days. Aggregation was monitored by a blue shift of the emission maximum (~ 354 nm for soluble tau, ~ 340 nm for aggregates).

Filter Trapping Assay—Aggregates of tau were trapped by filtration through a polyvinylidene difluoride membrane (pore diameter, 0.45 μm ; Schleicher & Schuell, Düren, Germany) adapted to a 96-well dot blot apparatus. The polyvinylidene difluoride membrane was wetted with methanol and rinsed with PBS buffer before insertion. The samples were pipetted into 100 μl of PBS and filtered. The membrane was washed three times with PBS before taking it out of the apparatus and blocked with 5% milk powder in PBS for 30 min in a rotational shaker at room temperature. The polyclonal antibody K9JA was used as primary antibody and incubated at a dilution of 1:20,000 at room temperature for 1 h. A secondary anti-rabbit antibody conjugated with horseradish peroxidase (Dako, Hamburg, Germany) was diluted 1:2000 and incubated for 30 min at 37 °C. After three times washing with TBS-Tween, the signal was detected using the ECL system (Amersham Biosciences). The pictures were taken with the digital gel documentation system Fuji film BAS3000 (Raytest, Straubenhardt, Germany) and quantified with the AIDA software package (Raytest, Straubenhardt, Germany).

Electron Microscopy—Protein solutions diluted to 0.1–10 μM were placed on 600-mesh carbon-coated copper grids for 1 min and negatively stained with 2% uranyl acetate for 45 s. The specimens were examined in a Philips CM12 electron microscope at 100 kV (Eindhoven, Netherlands).

Light Scattering Analysis of Tau-induced Microtubule Assembly—Microtubule assembly was monitored by light scattering at 350 nm in a Tecan spectrophotometer model Safire (Tecan, Crailsheim, Germany). Tau protein (10 μM) was mixed with tubulin dimer (30 μM) and GTP (1 mM) at 4 °C in polymerization buffer (100 mM Na-PIPES, pH 6.9, 1 mM EGTA, 1 mM MgSO_4 , 1 mM dithiothreitol) with a final volume of 40 μl . Tau and inhibitor compounds (60 μM) were added last. After rapid mixing, the samples were pipetted into a Greiner transparent flat bottom 384-well plate (4-mm path length) prewarmed at 37 °C. The reaction was started by incubating the cooled reaction components at 37 °C. The assembly of tubulin into microtubules was monitored by a change in turbidity. Three parameters were extracted from curves. The maximum turbidity at steady state, the rate of assembly, and the lag time between the temperature jump and the start of the turbidity rise.

Inducible Expression of Tau in N2a Cells—The generation of the cell model is described in detail elsewhere. Briefly, tau construct K18 Δ K280

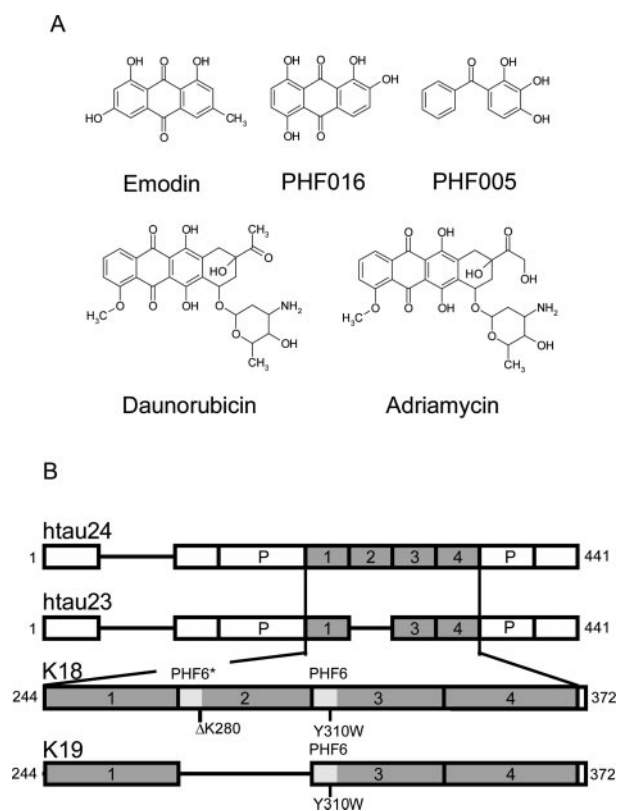


FIG. 1. Structure of inhibitor compounds, tau isoforms, and constructs. A, inhibitor compounds: emodin (1,3,8-trihydroxy-6-methyl-anthraquinone); PHF016 (1,2,5,8-tetrahydroxy-anthraquinone); PHF005 (1-phenyl-1-(2,3,4-trihydroxy-phenyl)-methanone); daunorubicin (8-acetyl-10-(4-amino-5-hydroxy-6-methyl-tetrahydro-pyran-2-yloxy)-6,8,11-trihydroxy-1-methoxy-7,8,9,10-tetrahydro-naphthacene-5,12-dione); and adriamycin (10-(4-amino-5-hydroxy-6-methyl-tetrahydro-pyran-2-yloxy)-6,8,11-trihydroxy-8-(2-hydroxy-ethanoyl)-1-methoxy-7,8,9,10-tetrahydro-naphthacene-5,12-dione). B, tau isoforms and constructs: htau24, a four-repeat isoform of tau lacking the two N-terminal inserts (numbering of the amino acids according to the longest isoform htau40); htau23, the fetal three-repeat isoform lacking the two N-terminal repeats and the second repeat (exon 10); construct K18 comprising the four repeats in the microtubule-binding domain; and construct K19 containing three repeats. The hexapeptide motifs PHF6 (third repeat) and PHF6* (second repeat) that promote the formation of β -structure are highlighted. The position of the point mutation Y310W in the third repeat is indicated.

was expressed in the mouse neuroblastoma cell line N2a in an inducible manner under the control of the reverse tetracycline-controlled transactivator (23). The cells were cultured in minimum essential medium with 10% fetal calf serum, 2 mM glutamine, and 0.1% nonessential amino acids. Expression of K18 Δ K280 was induced by adding 1 μg of doxycycline/1 ml of medium. The induction was continued over 7 days, and the medium was changed three times in 2-day intervals, complemented with doxycycline or with doxycycline plus tau aggregation inhibitors.

Assays of Tau Aggregation in N2a Cells—The levels and solubility of the K18 Δ K280 tau protein were determined by the method of Greenberg and Davies (24), which makes use of the insolubility of protein aggregates in cell homogenates after treatment with sarkosyl. The supernatant and sarkosyl-insoluble pellets were analyzed by Western blotting with the pan-tau antibody K9JA and analyzed by densitometry. Aggregation of tau in cells was tested by the fluorescence of thioflavin S. ThS signals were scored in three independent fields containing 40 cells each.

RESULTS

Compounds Sharing a Core Structure of Hydroxyanthraquinone Inhibit PHF Assembly—We tested 200,000 compounds in a high throughput screen for their capability of interfering with the aggregation of tau protein. Several members of the class of hydroxyanthraquinone and related structures were found (Fig. 1A). They all share a tricyclic aromatic ring system with some modifications. Emodin and compound PHF016 belong to the

class of hydroxyanthraquinones, whereas compound PHF005 represents a benzophenone derivative. Daunorubicin and adriamycin are enlarged with a further ring system to the naphthacendione structure and a sugar moiety.

The initial screening for inhibition of PHF aggregation was performed with the three-repeat tau construct K19 (Fig. 1B). This was chosen because it aggregates reliably at low micromolar concentrations in the presence of the polyanionic cofactor heparin and resembles Alzheimer's PHFs in terms of fiber morphology and β -sheet content (19, 25). The results described here were obtained by measuring the extent of aggregation via the fluorescence of ThS (22). This is based on the binding of ThS to fibers containing β -structure, which causes a change in the fluorescence spectrum (29). In the screen we performed polymerization of K19 in the presence and absence of inhibitors. Hits were defined by a >90% decrease of the ThS signal compared with the control reaction without compounds. This level of inhibition was reached by 1266 compounds (0.6% of the cohort). These were further tested for their ability to dissolve preformed PHFs, a more stringent criterion. 77 compounds were able to reduce the PHF-specific signal ThS fluorescence by more than 80%. To confirm the data we also used the four-repeat constructs K18 and K18 Δ K280 and the related full-length isoforms htau23 (three repeats, no inserts) and htau24 (four repeats, no inserts).

The five compounds discussed here are able to inhibit the transition from soluble to aggregated K19 protein, but they differ in their efficiency. Using fixed protein concentrations of K19 at 10 μ M, we tested the compounds in a concentration range from 10 pM to 200 μ M (Fig. 2A) and determined IC₅₀ values (Table I). Inhibitory effects begin to appear at ~0.1 μ M (ratio of protein to compound = 100) and reach nearly complete inhibition at 100 μ M compound (ratio protein to compound = 0.1). The curves of Fig. 2A decay fairly steeply over a compound concentration range of 2–3 orders of magnitude. The values of half-maximal inhibition (IC₅₀) are in the low micromolar range, which means that all compounds interfere with PHF aggregation of K19 already at substoichiometric concentrations. The four repeat construct K18 was tested under the same conditions (Fig. 2B), yielding IC₅₀ values down to the submicromolar range. However, the decay of the curves of K18 is more gradual than those of K19, extending over 3–4 orders of magnitude (compare Fig. 2A). This suggests that the filaments made from the three-repeat construct K19 are more stable and more homogeneous than those made from the four-repeat K18, possibly because K18 contains two nucleating hexapeptide motifs.

The study was then extended to the natural three- and four-repeat isoforms htau23 and htau24 (Fig. 1B). PHF formation of these proteins was assayed in the presence of 0.1, 1, 10, and 60 μ M compound (Fig. 2, C and D). Again a clear dose-dependent inhibition was observed. In this case the filaments from the three-repeat isoform htau23 are somewhat less stable than those from the four-repeat htau24. The compounds can be subdivided into two groups. For htau23 the more effective compounds are adriamycin, daunorubicin, and emodin, which are capable of inhibiting PHF formation ~50% at 0.1 μ M and ~90% at 60 μ M. Compounds PHF016 and PHF005 are less inhibitory, and they show only a slight effect at low concentration and a moderate one (~60%) at 60 μ M. For htau24 the ranking is roughly the same, with PHF016 and PHF005 showing little inhibitory activity.

All of the polymerization reactions carried out so far used heparin as a cofactor for inducing PHF assembly because otherwise the process would be impracticably slow (26, 27). To rule out a potential influence of heparin on the efficiency of the compounds, we used the four-repeat construct K18 Δ K280,

which carries one of the mutations observed in frontotemporal dementia (28) and is capable of aggregating into PHFs without a polyanionic cofactor (25) (Fig. 2, E and F). Overall the IC₅₀ values of assembly inhibition are higher or equal for K18 Δ K280 than for K18wt (suggesting that K18wt is more easily disrupted than the mutant), but the values are comparable with or without heparin, indicating that heparin does not greatly influence the interference between the compounds and PHFs.

To exclude a possible distortion of the data by the ThS dye, we wanted to confirm the results by a tryptophan fluorescence assay (30). It allows the detection of the molecular environment of a tryptophan introduced instead of tyrosine 310 whose emission maximum is sensitive to the burial in a more hydrophobic surrounding upon PHF formation. We therefore created the mutants K19/Y310W, K18/Y310W, and K18 Δ K280/Y310W (Fig. 1B) that contain a single tryptophan within the core of the PHF structure. In the soluble protein the emission maximum lies at ~354 nm (*dashed line*), whereas it shifts to 340 nm upon PHF formation (*dotted line*) (Fig. 3A, *first and second entries*). The emission peak can be shifted back by incubation at high concentrations of guanidine HCl, which is due to the disassembly of the PHFs (Fig. 3A, *fourth entry*).

In the Trp fluorescence assay the inhibition of PHF assembly becomes apparent if the emission maximum of Trp³¹⁰ remains higher than that of the control without any compound, because Trp³¹⁰ remains in a more solvent-accessible environment. The three repeat tau construct K19/Y310W (at 10 μ M) is prevented from aggregation by ~90% by all compounds at 60 μ M (Fig. 3A; note that the *fifth* through the *ninth entries* retain their values of ~354 nm, similar to control 1). By contrast the four repeat tau construct K18/Y310W is inhibited to this high extent only by PHF005 (Fig. 3B, *ninth entry*). Emodin, daunorubicin, and adriamycin can prevent PHF formation to ~70% at 60 μ M (Fig. 3B, *fifth, sixth, and seventh entries*), whereas PHF016 achieves only 25% inhibition (*eighth entry*). The trend becomes even more pronounced in the case of K18 Δ K280/Y310W, where all compounds show a lower activity (Fig. 3C). The internal ranking stays roughly the same as with K18; PHF005 (*ninth entry*) is the best, PHF016 (*eighth entry*) the worst inhibitor. Emodin, daunorubicin, and adriamycin (*fifth, sixth, and seventh entries*) showed a level of ~30–50% inhibition. The apparent degrees of inhibition differ somewhat between the ThS fluorescence and the intrinsic Trp fluorescence assays. One likely explanation is the different origin of the signal. In the ThS assay the dye has to bind to the filaments, which requires a minimal length of the fibers. The tryptophan fluorescence assay depends on the local surrounding of the residue and is therefore less dependent on the length of the filaments.

PHFs Can Be Dissolved by Inhibitor Compounds—Inhibiting the aggregation is one important aspect of a potential therapeutic compound, but even more important may be the ability to dissolve preformed filaments. We used the ThS assay to analyze this aspect with PHFs made from the repeat domain constructs K19 and K18 as well as from isoforms htau23 and htau24, containing three or four repeats. The disassembly of K19 filaments (Fig. 2G) follows similar concentration dependence as the inhibition experiment, with similar or slightly higher DC₅₀ values than the corresponding IC₅₀ concentrations (low micromolar range; Table I). K18 filaments show equal or slightly higher DC₅₀ values, *i.e.* the four-repeat constructs are more resistant to the compounds (Fig. 2H). Here, too, the concentration dependence for K19 is steeper than for K18 (Fig. 2, compare G and H), similar to that of assembly inhibition (Fig. 2, A and B), arguing for more heterogeneity in the K18 filaments.

PHFs from the full-length three-repeat isoform htau23 can

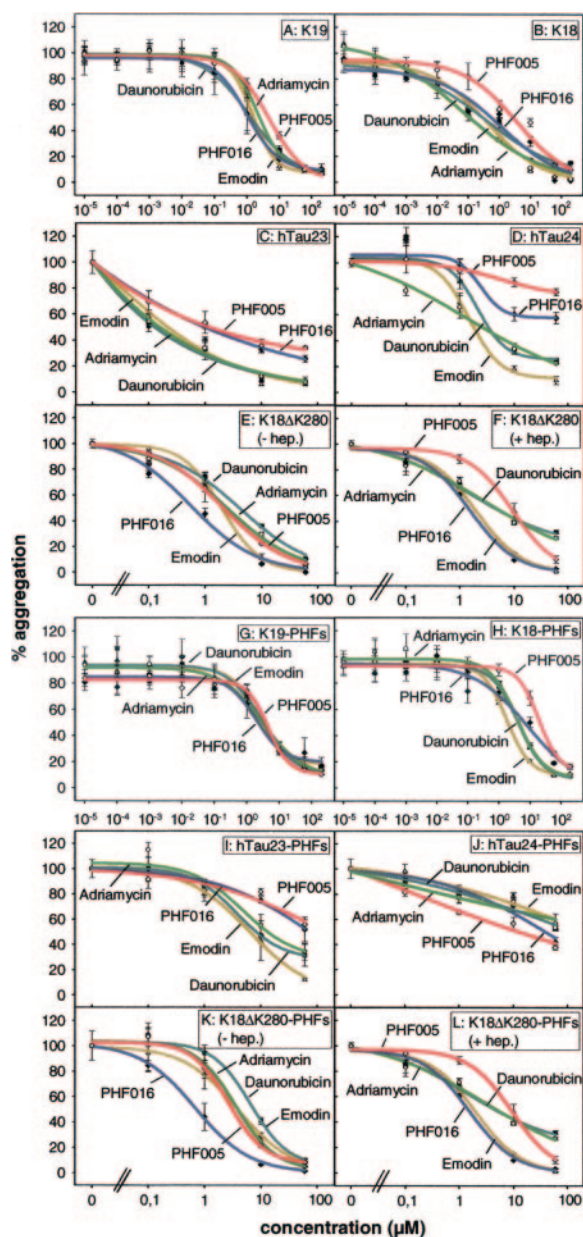


FIG. 2. Inhibition of PHF formation and disassembly of pre-formed PHFs induced by inhibitor compounds monitored by ThS fluorescence. A, aggregation of tau construct K19 (10 μ M) versus inhibitor concentration (10 pM to 200 μ M). The extent of aggregation was measured by the thioflavin S fluorescence assay, and the degree of inhibition was plotted as a percentage of control. All of the measurements were performed in triplicate. The compounds are color coded here and in the subsequent figures as follows: adriamycin (green), daunorubicin (cyan), emodin (dark yellow), PHF016 (blue), and PHF005 (red). The fits were calculated as four parameter logistic curves, and the IC_{50} values are summarized in Table I. Half-maximal inhibition occurs in the range of 1–5 μ M. B, inhibition of aggregation of construct K18. C, isoform hTau23. D, isoform hTau24. E, construct K18 Δ K280 without heparin. F, construct K18 Δ K280 with heparin. For disassembly experiments tau constructs and isoforms K19, K18, hTau23, and hTau24 (10 μ M) were first aggregated into PHFs as described, and the polymers were separated from the soluble tau by centrifugation of 1 h at 100,000 $\times g$, resolved, and then exposed to the inhibitors overnight at 37 $^{\circ}$ C at the indicated concentrations (range, 0.1–200 μ M). The compounds are capable of disassembling PHFs with varying efficiencies (see Table I). G, construct K19. H, K18. I, hTau23. J, hTau24. K, K18 Δ K280 (no heparin). L, K18 Δ K280 (with heparin). All of the measurements were performed in triplicate.

also be disassembled, albeit with lower efficiency (Fig. 2I). Emodin, daunorubicin, and adriamycin (DC_{50} range, 7.0–13 μ M) are more potent than PHF016 and PHF005 (DC_{50} > 60

μ M), consistent with the ranking in the assembly inhibition assay (Fig. 2C). By contrast, in the case of four-repeat htau24, all compounds have a low efficiency (Fig. 2J), the best values are achieved for PHF016 and PHF005 (DC_{50} values in the 10–40 μ M range). Thus, four-repeat full-length tau appears to be more stable than three-repeat tau, both in the assembly inhibition and in the disassembly assay.

The influence of heparin was tested by comparing the disassembly of filaments made from K18 Δ K280 without or with heparin (Fig. 2, K and L, and Table I). As in the assembly inhibition assay, the results are broadly comparable, showing that heparin does not have a major influence on the results (DC_{50} values range from 1 to 7.5 μ M).

As controls we performed the disassembly experiments with the tryptophan assay (Fig. 3, D–F) at compound concentrations of 60 μ M. In the tryptophan assay the compounds are able to dissolve K19 filaments (Fig. 3D), except daunorubicin (Fig. 3D, sixth entry). All other compounds yield emission maxima after treatment at $\sim$ 350–353 nm, close to the value of soluble tau, indicating a disassembly efficiency of $\sim$ 80–90%. In the case of K18 filaments (Fig. 3E), all compounds show a significantly lower efficiency of depolymerization; only PHF005 is a strong inhibitor under these conditions (80%), whereas emodin, adriamycin, and PHF016 exhibit not more than 10% efficiency. This ranking is consistent with the assembly inhibition assay (Fig. 3B). In the case of K18 Δ K280 (Fig. 3F), the efficiency of disassembly is further reduced, but the ranking remains comparable with that of K18 (compare Fig. 3E), as well as to the assembly inhibition assay (Fig. 3C). In these cases, PHF005 remains the most potent agent for depolymerizing PHFs (ninth entry).

As a third method we used a filter trap assay (Fig. 4). This assay monitors aggregated tau trapped on a membrane filter, whereas soluble protein is washed through. Therefore the technique preferably detects larger filaments, similar to the ThS assay. The compounds show similar disassembly activities as the ThS assay; emodin was most effective with a DC_{50} of $\sim$ 0.5 μ M.

All of the results were confirmed by electron microscopy. Fig. 5 illustrates examples of PHFs in the process of disassembly after overnight incubation with inhibitors. The filaments are seen in various stages of shortening or breaking, and the background contains numerous aggregates of undefined structure.

The general rule emerging from these experiments is that the compounds are more efficient in inhibiting *de novo* formation of PHFs than in depolymerizing preformed ones. This is revealed by the ThS fluorescence assay, by the filter trap assay, by the tryptophan fluorescence shift assay, and by electron microscopy and holds for all tau constructs and isoforms.

Submicromolar Concentrations Dissolve PHFs at Prolonged Incubation Times—For a potential therapeutic use one would like to have compounds that show activity both in inhibiting the *de novo* aggregation as well as in depolymerizing preexisting aggregates at submicromolar concentrations. Our depolymerization data were typically obtained after 12 h of incubation, but we were also interested in the effects of longer incubation times and lower compound concentrations. Fig. 6 shows the time course of disassembly (by the ThS assay) of K19 PHFs in the presence of 0.5 μ M adriamycin or PHF005 during 28 days. Nearly no effects are seen after 12 h, consistent with the earlier experiments (Fig. 2G), but interestingly the depolymerization still continues and results in a final depolymerization of $\sim$ 20–30% after 28 days. This suggests that even low concentrations of inhibitors can be used for disassembly at prolonged incubation times.

Aggregation Inhibitors Do Not Interfere with Tau-Microtubule Binding—The repeat domain of tau is not only important for PHF aggregation but also for the physiological function of

TABLE I
*IC*₅₀ values for PHF assembly inhibition and disassembly

Higher values indicate higher resistance of PHFs against inhibitor compounds.

Compound	Inhibition of tau aggregation (<i>IC</i> ₅₀)						Disassembly of PHFs (<i>DC</i> ₅₀)					
	K19	K18	K18ΔK280		hTau23	hTau24	K19	K18	K18ΔK280		hTau23	hTau24
			With heparin	Without heparin					With heparin	Without heparin		
			<i>μM</i>						<i>μM</i>			
Emodin	1.6	0.3	2.7	1.9	0.2	1.8	2.8	2.0	1.8	3.9	7.0	>60
Daurorubicin	1.3	0.3	4.5	4.0	0.1	3.4	3.1	4.0	4.5	7.4	8.2	>60
Adriamycin	2.4	0.1	4.1	2.4	0.2	2.7	2.9	4.3	4.8	3.9	13.2	>60
PHF016	1.1	0.6	1.2	0.6	1.0	>60	2.2	7.3	1.5	0.7	>60	39.2
PHF005	4.6	2.7	8.8	2.1	1.1	>100	3.8	20.8	8.8	3.1	>60	10.8

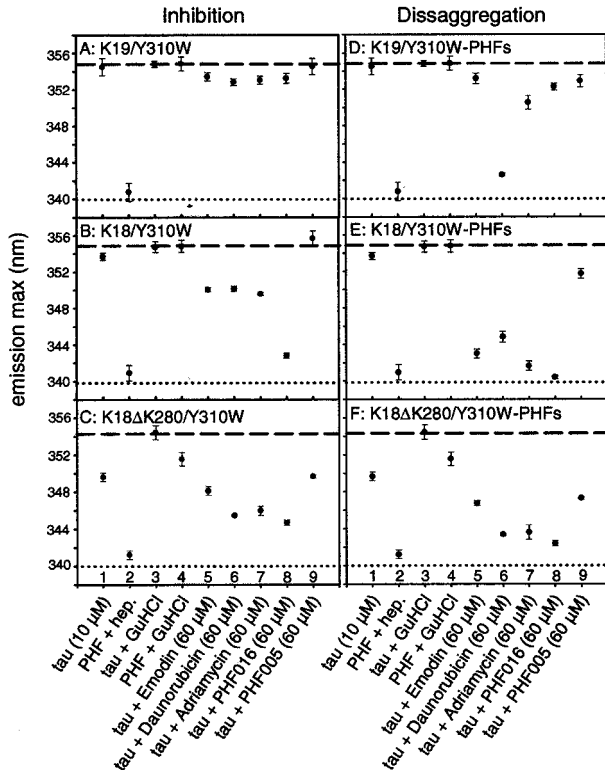


FIG. 3. Inhibition of PHF aggregation and disassembly of pre-formed PHFs induced by inhibitor compounds measured by tryptophan fluorescence shift assay. Fluorescence emission maximum of the single tryptophan Trp³¹⁰ inserted by site-directed mutagenesis into tau constructs K19 (A), K18 (B), and K18ΔK280 (C). Fully soluble tau constructs (10 *μM*) and tau or PHFs exposed to denaturing conditions (4 M guanidine HCl (*GuHCl*)) show the maximum of fully exposed Trp, aggregated PHFs show a maximum of 340 nm (typical of Trp buried in the interior), and tau aggregated in the presence of inhibitors (60 *μM*) shows intermediate values, depending on the degree of inhibition. Note that by this assay, all compounds are efficient inhibitors for the aggregation of the three-repeat construct K19/Y310W (Fig. 3A), but the four-repeat construct K18/Y310W (Fig. 3B) and its mutant K18ΔK280/Y310W (Fig. 3C) are much less responsive to the inhibitors. Disassembly of pre-formed PHFs made from K19/Y310W (D), K18/Y310W (E), and K18ΔK280/Y310W (F) induced by inhibitor compounds. Note that PHF aggregation is largely reversible for K19 (except for daunorubicin) but only partially for K18 and K18ΔK280. All of the measurements were done in triplicate.

microtubule binding. We therefore considered the possibility that the compounds inhibit both of these functions. Microtubule polymerization assays were performed with or without compounds (Fig. 7) in a modified assay format. Tubulin (at 30 *μM*) without tau serves as a negative control that is unable to self-assemble into microtubules because it is below the critical

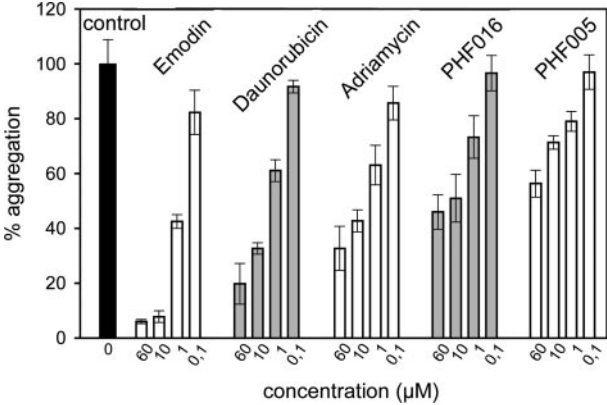


FIG. 4. Disassembly of pre-formed PHFs measured by filter assay. Depolymerization of PHFs from htau23 measured by filter assay. The bars show the fraction of polymerized material trapped on the polyvinylidene difluoride membrane. Black bar, control, untreated PHFs. The groups of bars show disassembly by emodin, daunorubicin, adriamycin, PHF016, and PHF005 as a function of compound concentration.

concentration. In the presence of tau (10 *μM*) tubulin polymerizes within 4 min. In the presence of compounds (60 *μM*) the rate and extent of polymerization are not significantly affected, except for daunorubicin. The same is true for Congo Red, an Aβ fiber inhibitor (31), used as a further control. These data suggest that the tested compounds influence specifically the pathological aggregation of tau protein but not its interaction with microtubules.

Cell Models of Tauopathy Can Be Rescued by Inhibitors—A crucial test for the inhibitors is their effect in cell models of tauopathy. To address this question we generated a neuroblastoma (N2a) cell line that allows inducible expression of the tau construct K18ΔK280 under the control of the tet-on transactivator. This construct was chosen because it contains the FTDP17 mutation ΔK280 in the four-repeat domain K18, which promotes the formation of β-structure and therefore aggregates readily *in vitro*, even without polyanionic inducers (25, 32). Expression of tau was initiated with 1 *μg/ml* doxycycline. The effect of aggregation inhibition was observed by adding the inhibitor emodin (15 *μM*). After 3–7 days the cells were tested for tau aggregation and thioflavin S fluorescence.

Fig. 8A shows Western blots of the cell extract after 7 days. Aggregated tau was separated from soluble tau by sarkosyl extraction, followed by pelleting. The pellet of the untreated control (–Emo) shows the typical “smear” at higher molecular weight that is characteristic of aggregation in Alzheimer’s disease (Fig. 8A, lane 2). However, emodin strongly suppresses the aggregates, leaving tau mostly in the soluble state (Fig. 8A, lane 4). Quantification of the sarkosyl-insoluble fraction shows a 5-fold reduction by emodin, from 14% of the total cellular tau down to 3% (Fig. 8B). Similar results were obtained by staining the cells with ThS (to show aggregated material) and with an

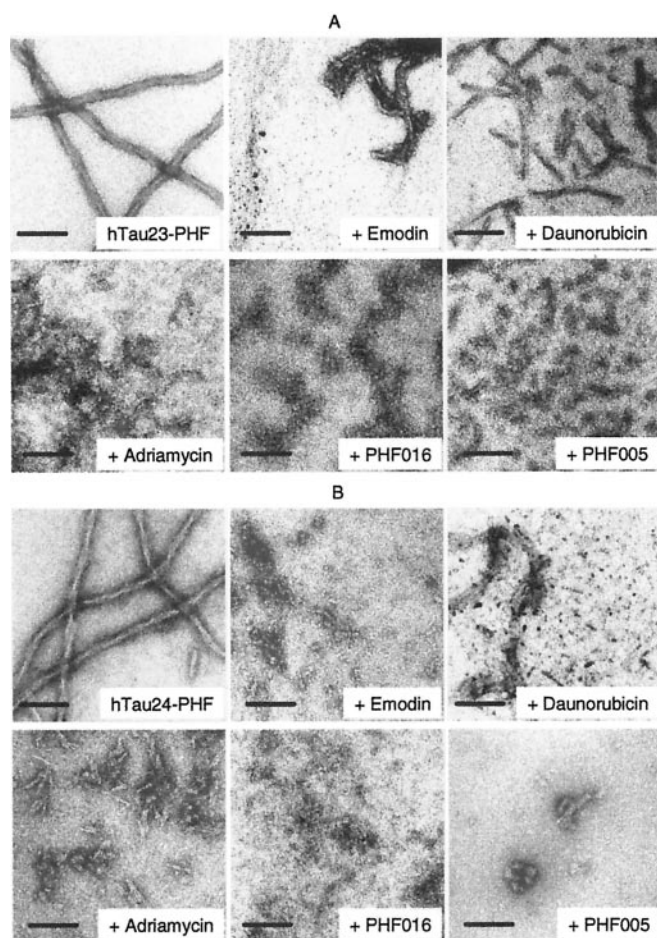


FIG. 5. **Electron microscopy of PHFs.** PHFs were assembled from htau23 (A, three repeats, no inserts) or htau24 (B, four repeats, no inserts) at $10\ \mu\text{M}$ protein concentration, $2.5\ \mu\text{M}$ heparin in PBS buffer, pH 7.4, 37°C . After assembly the inhibitor compounds were added at $60\ \mu\text{M}$, and the time course of disassembly was monitored by electron microscopy. The examples show breakdown products after overnight incubation with inhibitors. Bar, 100 nm.

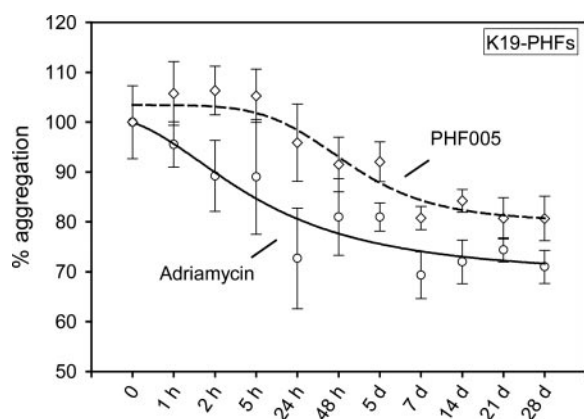


FIG. 6. **Time course of PHF disassembly (ThS assay) at low inhibitor concentrations.** PHFs were formed as above ($10\ \mu\text{M}$ construct K19, $2.5\ \mu\text{M}$ heparin, overnight) and then exposed to $0.5\ \mu\text{M}$ adriamycin (solid line) or PHF005 (dashed line). The degree of assembly was measured by the ThS assay. Note that despite the low inhibitor concentrations there is a gradual decrease of PHFs. Untreated controls were measured in parallel and subtracted as background.

antibody against total tau (to show the level of tau expression) (Fig. 8D). The level of tau expression was comparable without or with emodin (compare Fig. 8D, left column, top and bottom). However, whereas the ThS signal was strong in the tau ex-

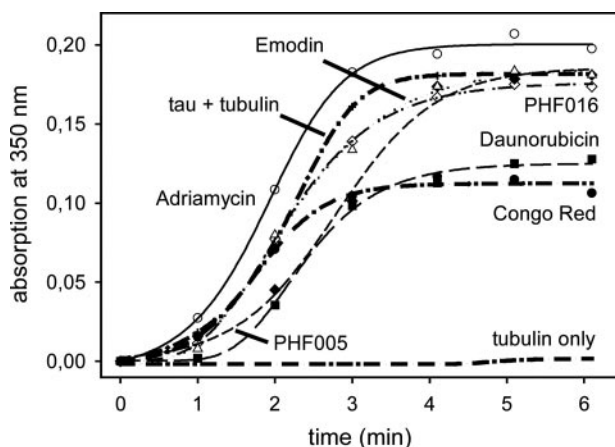


FIG. 7. **Effect of compounds on microtubule binding.** $30\ \mu\text{M}$ tubulin dimer was incubated in a microtiter plate at 37°C in the absence and presence of htau40 ($10\ \mu\text{M}$) and $60\ \mu\text{M}$ compound. Absorbance was taken at 350 nm and plotted versus time. The compounds are coded here as follows: adriamycin (solid line), daunorubicin (long dash line), emodin (dotted line), PHF016 (dash-dot-dot line), and PHF005 (short dash line). All of the curves (except tubulin only) show microtubule assembly within a few minutes.

pressing cells, it became very weak in the presence of emodin, consistent with the absence of aggregates (Fig. 8D, middle column, top and bottom). There were fewer ThS responsive cells, and fluorescence intensity was much lower as well. The merged images illustrate that a large fraction of cells contain visible aggregates (green-yellow in superposition), whereas the ThS signal was hardly visible in the emodin-treated cells (Fig. 8D, right column; quantification in Fig. 8C). These experiments show that tau aggregation inhibitors have the potential to reduce the level of aggregation without affecting the expression of tau as such and that they are able to rescue cells from the toxic effects of tau aggregates.

DISCUSSION

The anomalous aggregation of proteins into amyloid fibers is common to diverse human diseases, often related to advanced age. Factors contributing to this accumulation include the increased rate of protein misfolding, generation of amyloidogenic peptides, underactivity of repair systems (chaperones) and waste disposal (ubiquitin-proteasome system), or loss of energy supply and antioxidant defense mechanisms (impairment of mitochondria and peroxisome functions). There is a debate on whether fully aggregated proteins or smaller precursors (oligomers) cause cellular damage, but emerging evidence indicates that lowering the production or removal of the aggregates is one pathway toward alleviating the disease state (33–35). It is thus of interest to understand the pathway of aggregation and to identify and optimize inhibitors that might be of therapeutic value.

Despite the differences between the disease-causing proteins, the aggregation is often based on a similar principle, the formation of cross- β -structure where extended polypeptide segments embedded in β -sheets are aligned perpendicularly to the filament axis. The sheets provide binding surfaces for various dyes (Congo Red and thioflavin S). The conditions for cross- β -aggregation have been studied for various short peptides (36–41). Important parameters include (a) the conformation of the polypeptide as an extended chain capable of forming hydrogen bonds with parallel or antiparallel neighboring chains, (b) apolar or aromatic side chains at certain positions that promote the stacking of β -sheets by hydrophobic interactions, and (c) charged residues that promote electrostatic interactions and in addition provide guidance for the regular buildup of filaments.

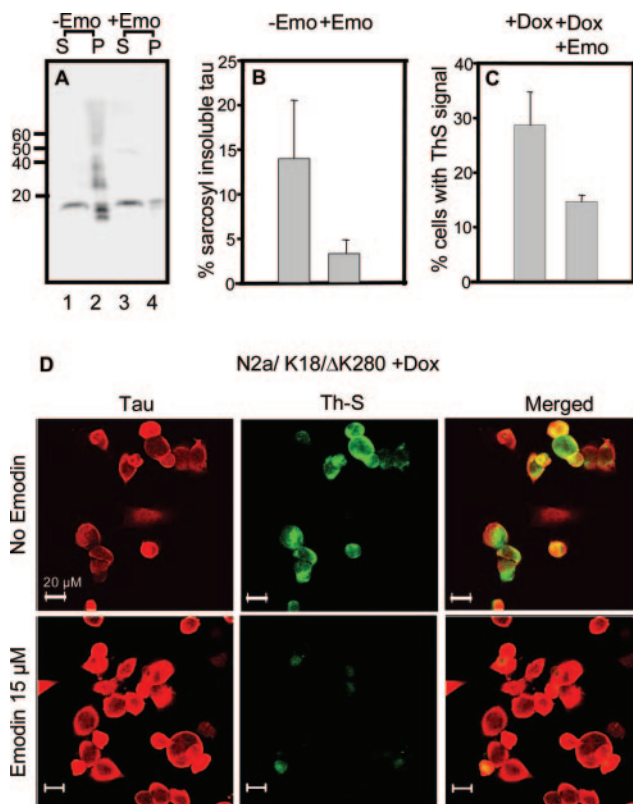


FIG. 8. Effect of the aggregation inhibitor emodin on tau aggregation in cells. A, Western blotting of fractionated lysates from inducible N2a cells expressing tau (K18 Δ K280) after sarkosyl extraction. Sarkosyl-insoluble K18 Δ K280 tau was detected in these cells after 7 days of induction. The sarkosyl-soluble (S) and insoluble pellet fractions (P) were separated by high speed centrifugation. The pellets obtained from cells incubated without (–) and with 15 μ M emodin (+) were resuspended in Tris-EDTA buffer in a volume equivalent to 5% of the extracts. Note that the amount of material loaded for supernatant and pellet represents 1 and 20% of the total extracted material, respectively. B, histogram of sarkosyl-insoluble tau (K18 Δ K280) from cells grown without emodin or with 15 μ M emodin (A, lanes 2 and 4). C, histogram of number of N2a cells expressing K18 Δ K280 (after induction with doxycyclin) with distinct thioflavin S signal in cell cultures induced without emodin (+Dox) or with 15 μ M emodin (+Dox, +Emo). Note that emodin inhibits the aggregation ~2-fold as measured by ThS. D, N2a cells were induced to express K18 Δ K280 and fixed after 3 days. They were sequentially double stained with pan-tau antibody K9JA (red) and thioflavin S (green). Top row, without emodin; bottom row, with 15 μ M emodin. Left column, immunofluorescence with tau antibody; middle column, ThS staining; right column, merge. Note the reduced ThS staining of cells in the presence of 15 μ M emodin (middle column, top and bottom).

By implication, the aggregation of filaments can be inhibited by inserting β -breaking residues such as proline into the sequence by disrupting the hydrophobic glue between sheets (e.g. mutation, oxidation, and phosphorylation of residues), by neutralizing Coulomb interactions (resulting in undirected amorphous aggregation rather than filaments). These considerations apply to the β -forming core sequence; in the case of larger proteins, where only a small fraction of the sequence is engaged in the cross- β -core, the remainder of the protein will determine whether the core peptide becomes accessible for interaction with other core peptides (42, 43). Thus, conceptually one has to distinguish between aggregation inhibitors that target the cross- β -core directly or the conformation of the protein that determines whether it is aggregation-competent or not. The first type of inhibitor would be expected to cross-react with several types of amyloid, whereas inhibitors of the second type could be specific for one particular protein.

In Alzheimer's disease there are two types of anomalous

protein aggregates: β -amyloid fibers formed from the A β peptide and PHFs formed from tau protein. In the case of β -amyloid, the cross- β -structure of the fibers was recognized early on because it dominates the filament structure (44, 45). With tau the nature of the PHFs was less obvious because tau is mostly a "natively unfolded" protein. Only a small fraction is engaged in forming the cross- β -backbone of the PHF core whose signature tends to be masked in biophysical studies of PHFs. The cross- β -structure of PHFs is centered on one or two hexapeptide motifs in the repeat domain of tau (19). Therefore one would expect that compounds that disrupt β -structure might also inhibit tau aggregation. In addition, the core of the PHFs is formed from the repeat domain and is thus larger than the hexapeptide motifs. One would also expect other types of inhibitory compounds acting on different parts of the protein, thereby altering conformations or other properties. Such compounds would be interesting for a tau-specific inhibition of aggregation.

To identify inhibitors of PHF aggregation we screened a library of 200,000 compounds. Promising compounds were probed by secondary assays, such as sedimentation and electron microscopy, and the example of one subclass (anthracenes) is described here. We also developed a cell model of tau aggregation to probe the effects of tau aggregation inhibitors. We wanted to answer several questions: (a) What types of compounds can inhibit the aggregation of tau? These PHF inhibitor compounds could be useful in preventing aggregation and thus delaying the disease process. (b) Are these compounds also effective in disassembling preassembled PHFs? Such PHF breaker compounds might help to reverse the disease process. (c) Are compounds that inhibit tau aggregation *in vitro* also active in a cellular environment? (d) Is the inhibition of aggregation beneficial for cells? The last two questions are important in the light of the presumed functions of tau in cells. It is *a priori* not clear how the balance between aggregated and soluble tau would affect the physiology of cells. The best known function of tau is to stabilize axonal microtubules (which is considered positive for the cell and which is deficient in Alzheimer's disease). However, an increase in the level of tau can impair axonal transport (46), and it can interfere with the dynamic instability of microtubules that is necessary for neuronal survival and axonal growth (47). These two properties of tau could, in principle, cause damage to cells independently of aggregation.

Many of the inhibitory compounds found in the initial screen contain ring structures. This might seem surprising, considering that tau is a hydrophilic molecule with very few aromatic residues. Part of the explanation is that ring structures are abundant in the compound library, but the likely structural interpretation is that these inhibitors interrupt the β -structure around the hexapeptide motifs of tau, for example by intercalating between the hydrophobic residues and thus disrupting the interface between two neighboring sheets. Other reported inhibitors of tau aggregation have ring structures as well, such as phenothiazines (48) or a benzothiazolidine derivative (49). We also note that some compounds that have shown up in our screen have been found in other screens designed for the inhibition of other amyloidogenic peptides like A β (e.g. daunomycin (50)) or transthyretin (e.g. a doxorubicin derivative (51)).

When considering the origins of tauopathies, the issue of phosphorylation deserves special attention. Because of its unfolded nature and numerous serines and threonines, the protein can be phosphorylated at many sites and by many kinases. Some sites cause the detachment of tau from microtubules, others may control interactions with partners (e.g. docking sites for kinases or phosphatases). In all tauopathies, tau is

hyperphosphorylated. Whether or not this promotes aggregation is a matter of debate, but if one makes that assumption it would be natural to reduce the activity of the responsible kinases (or enhance the activity of phosphatases). Thus, in current efforts to find inhibitors of tauopathy one can distinguish two approaches, screens for inhibitors of kinase signaling (52–54), and aggregation inhibitors (49). Our own experimental evidence suggests that phosphorylation *in vitro* does not promote tau aggregation but rather reduces it (6), and therefore we decided to search directly for aggregation inhibitors.

Finally, the crucial question is: Can the application of tau aggregation inhibitors prevent or rescue an organism from tauopathy? This issue can be addressed by using cell models of tauopathy. The development of suitable models has been slow because overexpression of tau *per se* does not lead to aggregation, because of the high solubility and phosphorylation of tau. However, following the discovery of tau mutations in FTDP-17, several cell models have been created that express tau mutants (54–56). In our cell model we have combined two features. One is the choice of the mutation $\Delta K280$, because it strongly promotes the aggregation of tau. The second is the inducible expression in a neuroblastoma cell line (N2a), because it allows one to ask whether tau aggregates emerge with time after induction and to check whether they disappear when the tau expression is switched off again. In the context of the present study, the important point is that tau aggregates can be made to disappear by treating the cells with an inhibitor. This suggests that pharmacological approaches to contain the aggregation of tau could be a valid approach for tauopathies such as Alzheimer's disease.

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